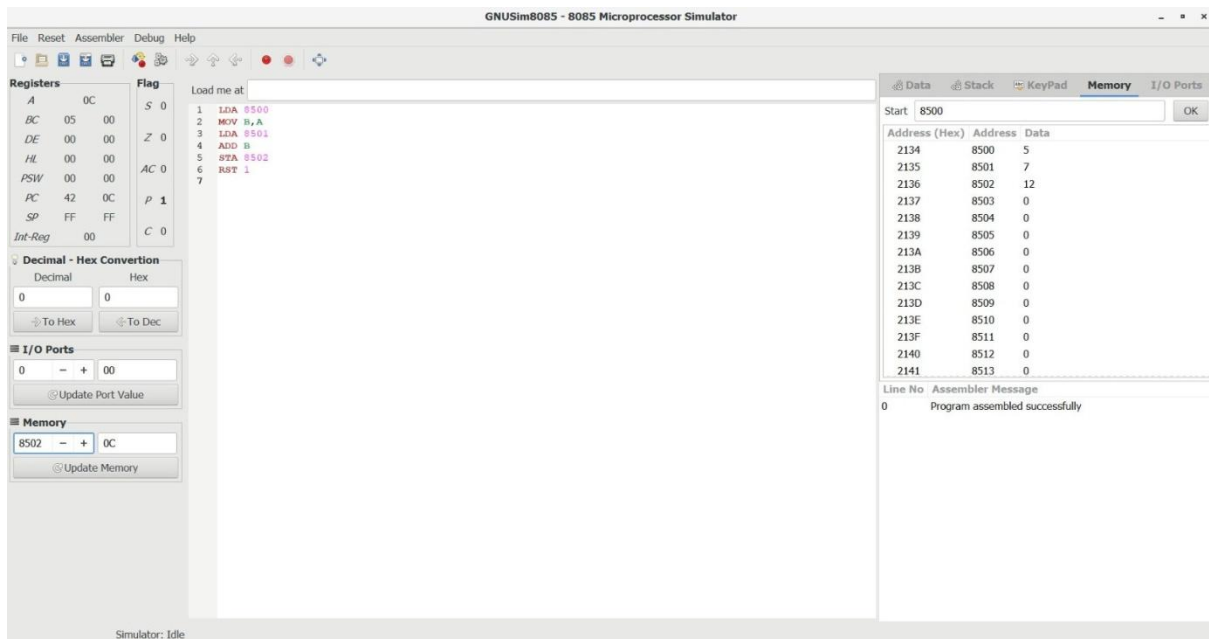
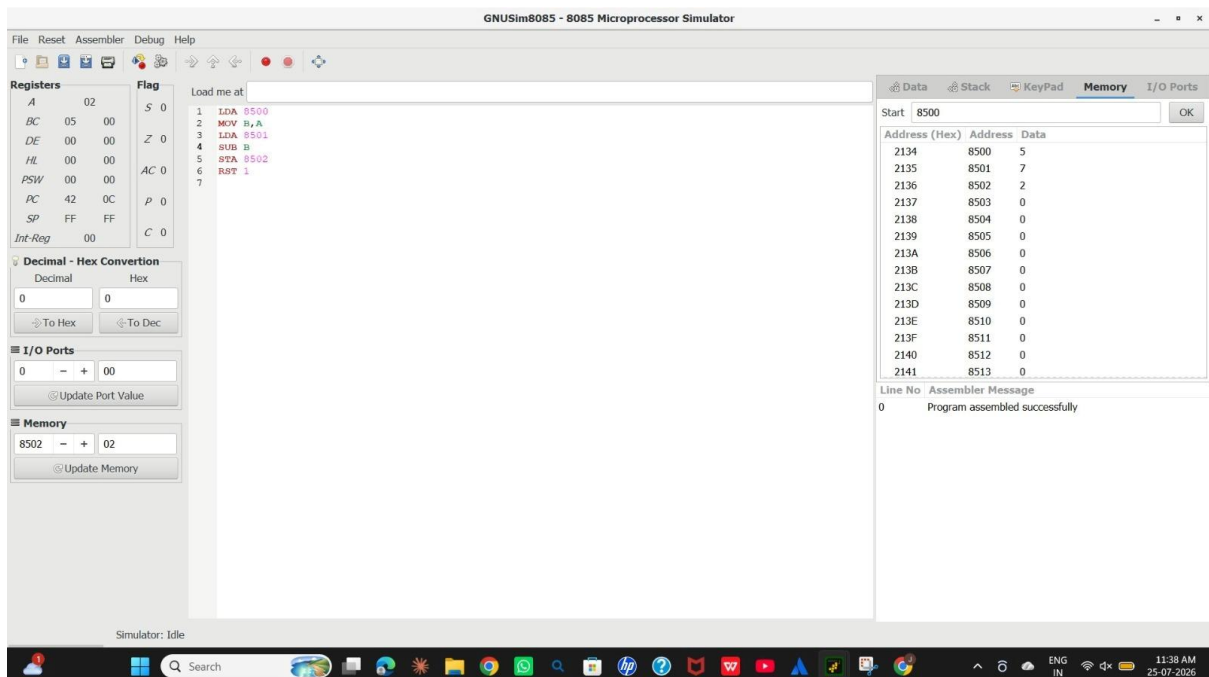


1) 8 Bit Addition



2) 8 Bit Subtraction



3) 8 Bit Add with carry

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for an 8-bit addition with carry. The registers and flags are shown on the left, and the memory dump is on the right.

Registers:

Register	Value
A	32
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

Flags:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Assembly Code:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MVI A,90
6 LOOP: ADD B
7 DCR C
8 JNZ LOOP
9 STA 2502
10 HLT
```

Memory Dump:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	5
09C6	2502	50
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message:

```
0 Program assembled successfully
```

4) 8 Bit Sub with Borrow

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for an 8-bit subtraction with borrow. The registers and flags are shown on the left, and the memory dump is on the right.

Registers:

Register	Value
A	05
BC	0A 02
DE	05 00
HL	00 00
PSW	00 00
PC	42 1C
SP	FF FF
Int-Reg	00

Flags:

Flag	Value
S	0
Z	1
AC	1
P	1
C	1

Assembly Code:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MVI D,90
6 MOV A,B
7 LOOP: SUB C
8 JC STOP
9 INR D
10 JMP LOOP
11 STOP: ADD C
12 STA 2503
13 MOV A,D
14 STA 2502
15 HLT
```

Memory Dump:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	2
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message:

```
0 Program assembled successfully
```

5) 8 Bit Multiplication

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the following components:

- Registers:** A table showing the state of registers. The PC register is 42, and the SP register is FF. The Int-Reg is 00.
- Flag:** A table showing the state of flags. The S flag is 0, and the Z flag is 0.
- Decimal - Hex Conversion:** A section with input fields for Decimal and Hex values, and buttons for conversion.
- I/O Ports:** A section with input fields for port values and buttons for updating.
- Memory:** A section with input fields for memory address and value, and buttons for updating.
- Assembly Code:** A list of instructions: 1 LDA 8500, 2 MOV B,A, 3 LDA 8501, 4 SUB B, 5 STA 8502, 6 RST 1, 7.
- Memory Window:** A table showing memory addresses and data. The Start address is 8500. The data at 8500 is 5, at 8501 is 7, and at 8502 is 2.
- Assembler Message:** A message box showing "Program assembled successfully".

The simulator status bar at the bottom indicates "Simulator: Idle".

6) Division with Quotient

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the following components:

- Registers:** A table showing the state of registers. The PC register is 42, and the SP register is FF. The Int-Reg is 00.
- Flag:** A table showing the state of flags. The S flag is 0, and the Z flag is 0.
- Decimal - Hex Conversion:** A section with input fields for Decimal and Hex values, and buttons for conversion.
- I/O Ports:** A section with input fields for port values and buttons for updating.
- Memory:** A section with input fields for memory address and value, and buttons for updating.
- Assembly Code:** A list of instructions: 1 LDA 2500, 2 MOV B,A, 3 LDA 2501, 4 CMP B, 5 JC FIRST, 6 STA 2502, 7 HLT, 8 FIRST: MOV A,B, 9 STA 2502, 10 HLT.
- Memory Window:** A table showing memory addresses and data. The Start address is 2500. The data at 2500 is 10, at 2501 is 20, and at 2502 is 20.
- Assembler Message:** A message box showing "Program assembled successfully".

The simulator status bar at the bottom indicates "Simulator: Idle".

7) Division without Quotient

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for a program that performs division without quotient. The code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

The Registers window shows the following values:

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0F
SP	FF FF
Int-Reg	00

The Flag window shows the following values:

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

The Memory window shows the following values:

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	20
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports window shows the following values:

Port	Value
0	00

The Assembly Message window shows the following message:

```
Line No: 0
Assembler Message: Program assembled successfully
```

8) Greatest Of 2 Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for a program that finds the greatest of two numbers. The code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC FIRST
6 STA 2502
7 HLT
8 FIRST: MOV A,B
9 STA 2502
10 HLT
```

The Registers window shows the following values:

Register	Value
A	14
BC	14 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The Flag window shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	0
C	1

The Memory window shows the following values:

Address (Hex)	Address	Data
09C4	2500	20
09C5	2501	15
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports window shows the following values:

Port	Value
0	00

The Assembly Message window shows the following message:

```
Line No: 0
Assembler Message: Program assembled successfully
```

9) Smallest Of 2 Numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 CMP B
5 JC SECOND_SMALLER
6 MOV A,B
7 STA 2502
8 HLT
9 SECOND_SMALLER: STA 2500
10 HLT
```

The registers section shows the following values:

Register	Value
A	05
BC	1E 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory section shows the following values:

Address (Hex)	Address	Data
09C4	2500	30
09C5	2501	5
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports section shows the following values:

Port	Value
0	00

The Memory section shows the following values:

Address	Value
2501	05

The simulator status is "Simulator: Idle".

10) Swapping

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MOV A,B
6 STA 2501
7 MOV A,C
8 STA 2500
9 HLT
```

The registers section shows the following values:

Register	Value
A	28
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 11
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

The memory section shows the following values:

Address (Hex)	Address	Data
09C4	2500	40
09C5	2501	50
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The I/O Ports section shows the following values:

Port	Value
0	00

The Memory section shows the following values:

Address	Value
2500	28

The simulator status is "Simulator: Idle".

11) 1s Complement

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The Registers window on the left displays the status of various registers and flags. The central assembly window shows a program with four instructions: LDA 2500, CMA, STA 2501, and HLT. The Memory window on the right shows the memory contents at addresses 09C4 to 09D1. The I/O Ports window shows the port value as 00. The Assembler Message window at the bottom indicates that the program was assembled successfully.

Registers

Register	Value
A	E3
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 08
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

Assembly

```
1 LDA 2500
2 CMA
3 STA 2501
4 HLT
```

Memory

Address (Hex)	Address	Data
09C4	2500	28
09C5	2501	227
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

I/O Ports

Port Value: 00

Assembler Message

```
Line No: 0
Assembler Message: Program assembled successfully
```

12) 2s Complement

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The Registers window on the left displays the status of various registers and flags. The central assembly window shows a program with nine instructions: LDA 2500, ANI 01H, JZ EVEN, MVI A, 01H, STA 2501, HLT, EVEN: MVI A, 00H, STA 2501, and HLT. The Memory window on the right shows the memory contents at addresses 09C4 to 09D1. The I/O Ports window shows the port value as 00. The Assembler Message window at the bottom indicates that the program was assembled successfully.

Registers

Register	Value
A	00
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 14
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	1
P	1
C	0

Assembly

```
1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT
```

Memory

Address (Hex)	Address	Data
09C4	2500	24
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

I/O Ports

Port Value: 00

Assembler Message

```
Line No: 0
Assembler Message: Program assembled successfully
```

13) Odd/Even

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	01
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 0E
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	1
P	0
C	0

Load me at

```
1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 23

Update Memory

Start: 2500

Address (Hex)	Address	Data
09C4	2500	35
09C5	2501	1
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

14) Positive/Negative

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	00
BC	32 28
DE	00 00
HL	00 00
PSW	00 00
PC	42 13
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	0
AC	0
P	0
C	0

Load me at

```
1 LDA 2500
2 ORA A
3 JP POSITIVE
4 MVI A, 01H
5 STA 2501
6 HLT
7 POSITIVE: MVI A, 00H
8 STA 2501
9 HLT
```

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

2500 - + 19

Update Memory

Start: 2500

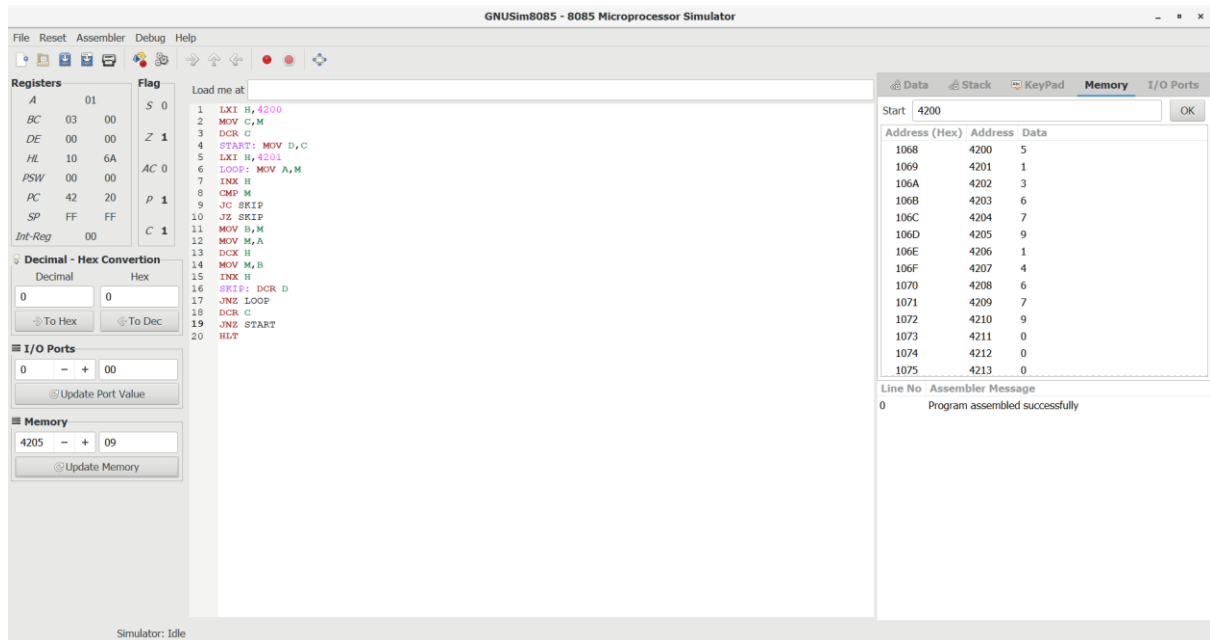
Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	0
09C6	2502	5
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

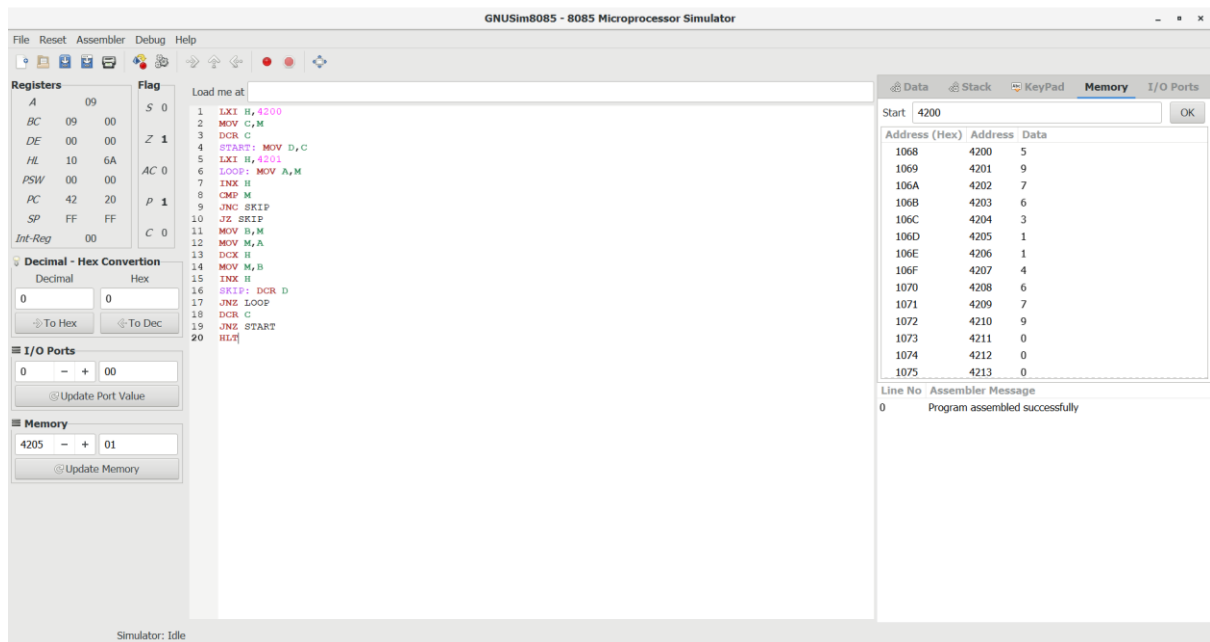
0 Program assembled successfully

Simulator: Idle

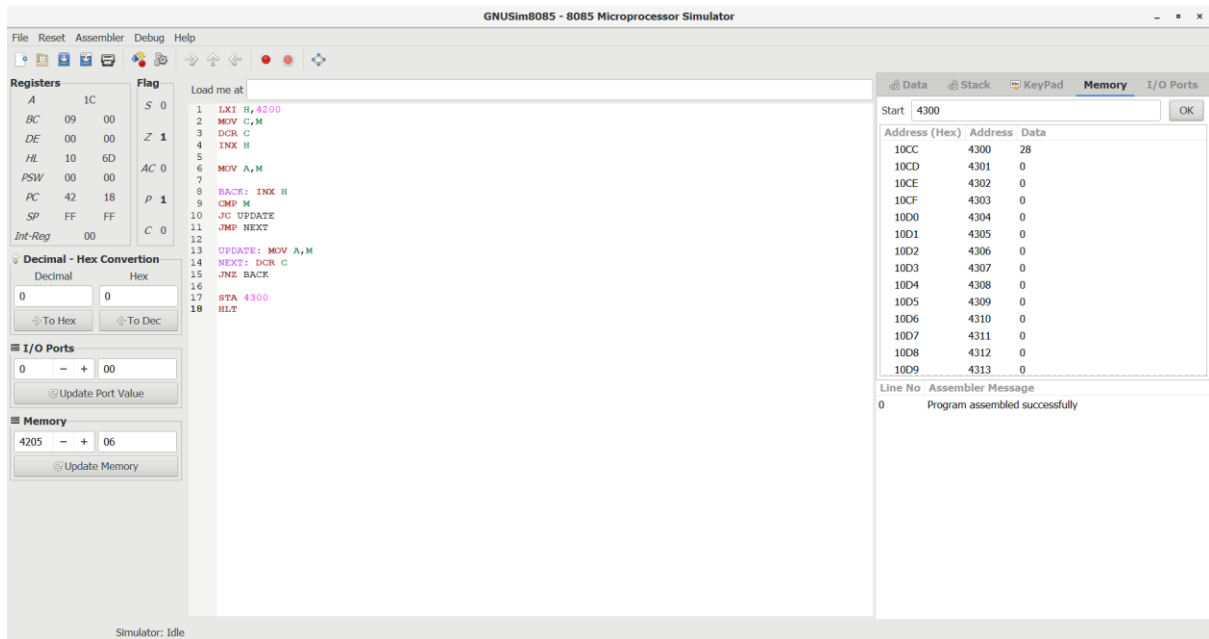
Write an assembly language program to arrange numbers in Ascending order.



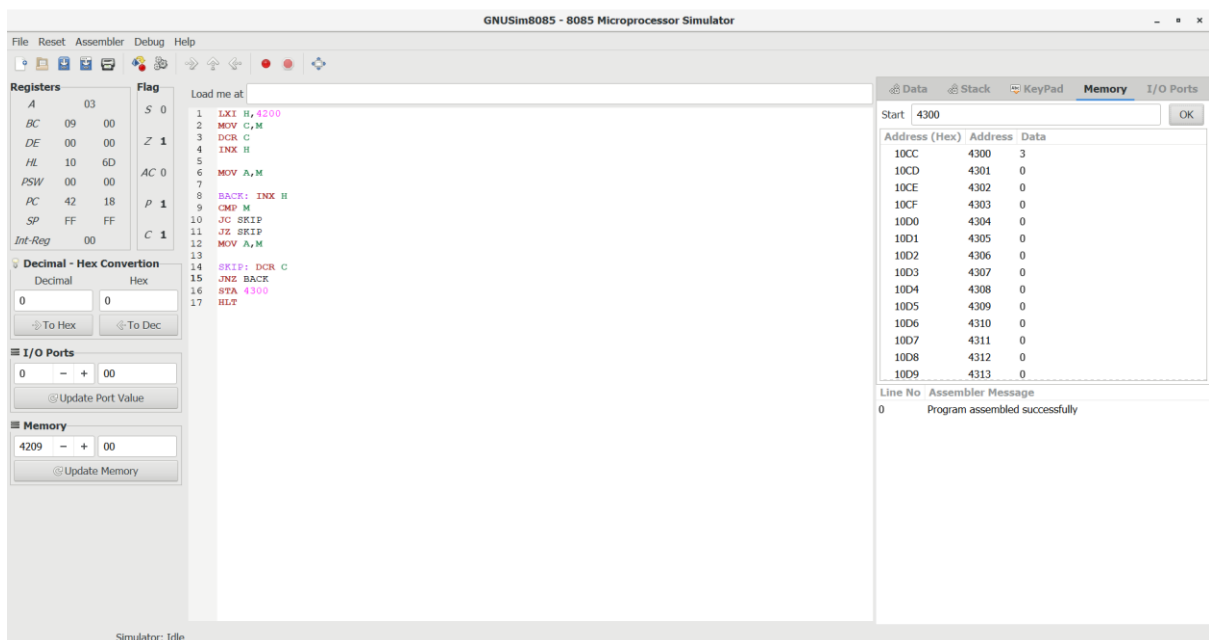
Write an assembly language program to arrange numbers in Descending order.



Write an assembly language program to find the largest number in an array.



Write an assembly language program to find the Minimum number in an array.



LCM of Two 8-bit Numbers using 8085

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for calculating the LCM of two 8-bit numbers. The code is as follows:

```
1 LDA 4200
2 STA 4300
3
4 LDA 4201
5 MOV E, A
6
7 CHECK: LDA 4300
8 MOV D, A
9 DIVIDE: MOV A, D
10 CMP E
11 JC NOT_DIVISIBLE
12 JZ FOUND
13 SUB E
14 MOV D, A
15 JMP DIVIDE
16 NOT_DIVISIBLE: LDA 4300
17 MOV B, A
18 LDA 4200
19 ADD B
20 STA 4300
21 JMP CHECK
22 FOUND: HLT
```

The registers section shows the following values:

Register	Value
A	02
BC	03 00
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory section shows the following values:

Address (Hex)	Address	Data
10CC	4300	6
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

The assembler message section shows the following message:

```
Line No Assembler Message
0 Program assembled successfully
```

GCD of Two 8-bit Numbers using 8085

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for calculating the GCD of two 8-bit numbers. The code is as follows:

```
1 LDA 4200
2 MOV B, A
3 LDA 4201
4 MOV C, A
5 LOOP: MOV A, B
6 ORA A
7 JZ ZERO
8 MOV A, C
9 ORA A
10 JZ ZERO
11 MOV A, B
12 CMP C
13 JC GREATER
14 JZ EQUAL
15 SUB C
16 MOV B, A
17 JMP LOOP
18 GREATER: MOV A, C
19 SUB B
20 MOV C, A
21 JMP LOOP
22 EQUAL: MOV A, B
23 STA 4205
24 HLT
25 ZERO: MOV A, C
26 STA 4205
27 HLT
28 ZERO: MOV A, B
29 STA 4205
30 HLT
```

The registers section shows the following values:

Register	Value
A	01
BC	01 01
DE	02 02
HL	00 00
PSW	00 00
PC	42 2A
SP	FF FF
Int-Reg	00

The flags section shows the following values:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

The memory section shows the following values:

Address (Hex)	Address	Data
1068	4200	3
1069	4201	2
106A	4202	0
106B	4203	0
106C	4204	0
106D	4205	1
106E	4206	0
106F	4207	0
1070	4208	0
1071	4209	0
1072	4210	0
1073	4211	0
1074	4212	0
1075	4213	0

The assembler message section shows the following message:

```
Line No Assembler Message
0 Program assembled successfully
```

Factorial of an 8-bit Number using 8085

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for calculating the factorial of an 8-bit number. The registers and flags are shown on the left, and the memory window on the right displays the program's execution memory.

Registers:

Register	Value
A	78
BC	78 00
DE	06 78
HL	00 00
PSW	00 00
PC	42 28
SP	FF FF
Int-Reg	00

Flags:

Flag	Value
S	1
Z	0
AC	0
P	1
C	1

Assembly Code:

```
1 LDA 4200
2 MOV C,A
3
4 MVI B,01
5 MVI D,02
6
7 CHECK: LDA 4200
8 CMP D
9 JC DONE
10 JZ MULTIPLY
11
12 MULTIPLY: MOV A,D
13 MOV C,A
14 MVI E,00
15
16 ADD_LOOP: MOV A,E
17 ADD B
18 MOV E,A
19 DCR C
20 JNZ ADD_LOOP
21
22 MOV A,E
23 MOV B,A
24
25 INR D
26 JMP CHECK
27
28 DONE: MOV A,B
29 STA 4300
30 HLT
```

Memory Window:

Address (Hex)	Address	Data
10CC	4300	120
10CD	4301	0
10CE	4302	0
10CF	4303	0
10D0	4304	0
10D1	4305	0
10D2	4306	0
10D3	4307	0
10D4	4308	0
10D5	4309	0
10D6	4310	0
10D7	4311	0
10D8	4312	0
10D9	4313	0

Assembler Message:

```
Line No Assembler Message
0 Program assembled successfully
```

Convert decimal into hexadecimal

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window displays the assembly code for converting a decimal number into hexadecimal. The registers and flags are shown on the left, and the memory window on the right displays the program's execution memory.

Registers:

Register	Value
A	A9
BC	90 09
DE	A0 00
HL	00 00
PSW	00 00
PC	42 20
SP	FF FF
Int-Reg	00

Flags:

Flag	Value
S	1
Z	0
AC	0
P	1
C	0

Assembly Code:

```
1 LDA 4200
2 MOV B,A
3
4 ANI 0FH
5 MOV C,A
6
7 MOV A,B
8 ANI 0FH
9
10 RRC
11 RRC
12 RRC
13 RRC
14
15 MOV B,A
16
17 MVI D,00
18 MVI E,0AH
19
20 TEN_TIMES: MOV A,D
21 ADD B
22 MOV D,A
23 DCR B
24 JNZ TEN_TIMES
25 MOV A,D
26 ADD C
27 STA 4205
28
29 HLT
```

Memory Window:

Address (Hex)	Address	Data
1068	4200	25
1069	4201	2
106A	4202	0
106B	4203	0
106C	4204	0
106D	4205	169
106E	4206	0
106F	4207	0
1070	4208	0
1071	4209	0
1072	4210	0
1073	4211	0
1074	4212	0
1075	4213	0

Assembler Message:

```
Line No Assembler Message
0 Program assembled successfully
```